Clinical Decision Support System for Disease Risk Assessment

Progress Report - 5

Submitted By: Uthara Uday (300395271)

Course: CSIS 4495 – Applied Research Project

Section: 03

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1. Work Date / Hours Log

Date	Hours	Description of Work Done
Mar 20, 2026	3.5	Redesigned dashboard layout into light theme with structured sections and improved visual hierarchy.
Mar 21, 2026	4.5	Implemented summary cards, health indicators display, and simplified patient view.
Mar 22, 2026	3.5	Integrated AI chat assistant using API key for patient-friendly explanations.
Mar 23, 2026	4.0	Developed Altair trend charts and resolved rendering errors in visualization.
Mar 23, 2026	3.0	Improved UI responsiveness, spacing, icons, and reduced text-heavy sections.

2. Summary Description of Work Done During This Reporting Period

In this reporting period, I was working on improving the user interface and interactivity of the application in sampleapp.py. The dashboard has been redesigned into a light themed design with better readability and user experience.

Some of the critical contributions were the introduction of summary cards, structured sections, simplified outputs in such a way that patients can easily comprehend their results. It was provided with an AI chat assistant that can be asked questions about their reports and provide them with simplified answers with the help of API keys.

Moreover, trend charts with the help of Altair were also introduced to represent the changes of the main health indicators over time. The refinements made on the UI, like the improved spacing, icons, and text density, were also introduced to make the application more interesting and user-friendly.

3. Repo Check-In of Implementation Completed

Updated File

- sampleapp.py

Updates

The sampleapp.py file was greatly improved to make the application more patient oriented dashboard than a data display interface. The new features were UI redesign, the introduction of an AI chat assistant, the introduction of trend charts to visualize indicators, and the reorganization of the content into clear and interactive blocks. These enhancements made it easier to use, interpret, and engage with the system.

4. AI Use Section

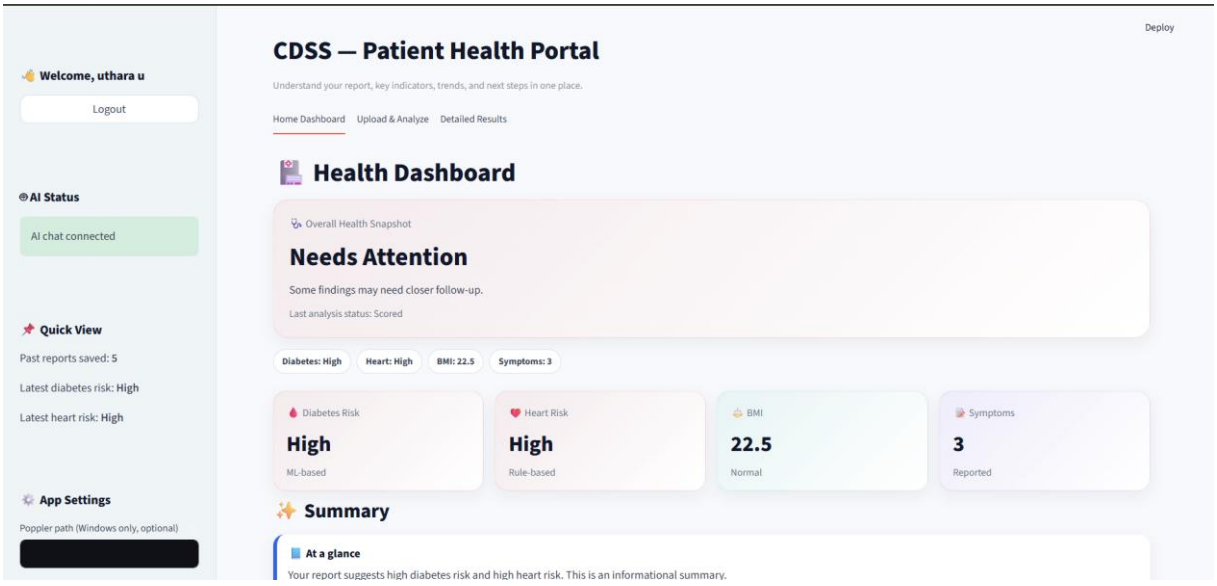
AI Tool	Version	Purpose of Use	Value Added by User
ChatGPT	GPT-5 (Premium)	UI redesign and debugging charts	Adapted suggestions into a structured and user-friendly dashboard
OpenAI API	Latest	AI chatbot integration	Designed prompts for safe, non-diagnostic explanations

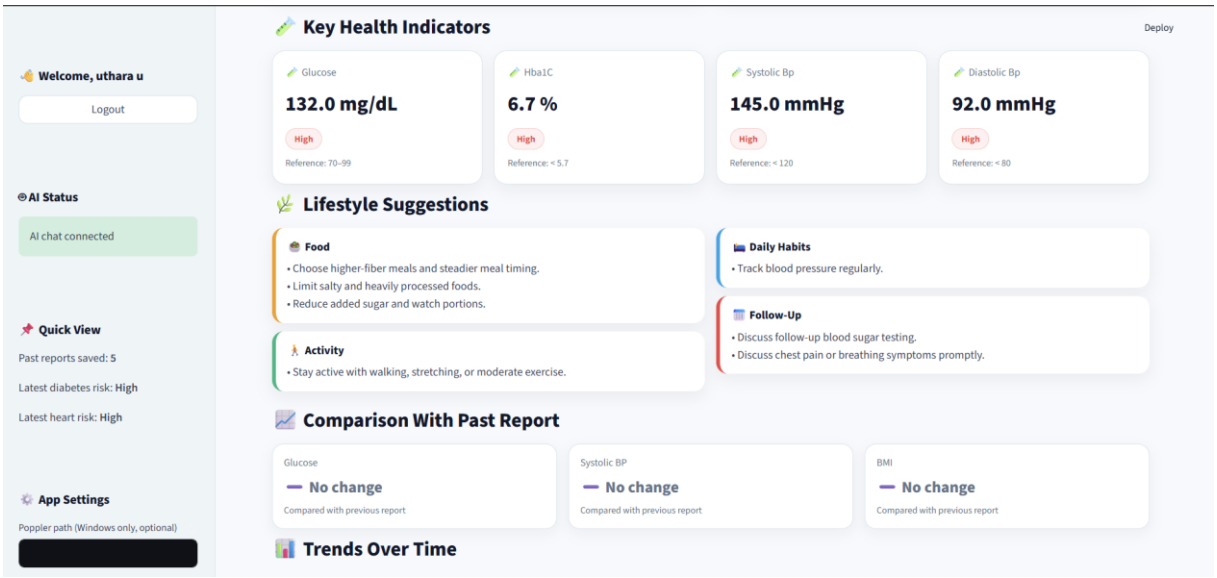
5. APPENDIX

5.1 Appendix A – AI Prompts

1. How to implement trend charts using Altair in Streamlit?

5.2 Appendix B – Screenshots





Welcome, uthara u

Logout

AI Status

AI chat connected

Quick View

Past reports saved: 5

Latest diabetes risk: High

Latest heart risk: High

App Settings

Poppler path (Windows only, optional)

Glucose

Systolic BP

BMI

Date

Deploy

Ask AI About This Report

what does my glucose mean

Your glucose level is higher than the usual range, which means your blood sugar is elevated. This can be a sign that your body is having trouble managing sugar properly. Since your glucose is high along with other factors like blood pressure and cholesterol, it may suggest a higher risk for conditions like diabetes or heart issues.

It's important to follow up with a healthcare provider to explore this further. They can help determine what's going on and suggest ways to manage your health. Meanwhile, staying active, eating balanced meals, and avoiding sugary foods can be helpful steps.

If you experience severe symptoms like chest pain or very short breath, seek medical help right away. This app provides information but is not a diagnosis.

Ask a question about the report

what does my glucose mean

Send

This assistant explains results in simple language and does not provide diagnosis.

Home Dashboard | Upload & Analyze | Detailed Results

Deploy

Upload Report & Run Analysis

Patient Profile

Age: 25

Height (cm): 170.00

Smoker: No

Sex: Prefer not to say

Weight (kg): 65.00

Family History - Diabetes: No

Family History - Heart Disease: No

Symptoms

- [ ] Fatigue
- [x] Increased thirst
- [ ] Chest pain
- [ ] Dizziness
- [ ] Frequent urination
- [ ] Blurred vision
- [x] Shortness of breath
- [x] Palpitations

BMI: 22.5 — Automatic

Category: Normal — Based on height and weight

Wellness: Stable — General view

Ready to upload

Upload a report to generate risk summary, trends, and AI explanation.

Upload Medical Report

Upload a PDF/JPG/PNG report

Drag and drop file here

Limit 200MB per file • PDF, JPG, JPEG, PNG

Choose a file to process.

Analysis Progress

Latest run status: Scored

Run Analysis

Home Dashboard | Upload & Analyze | Detailed Results

Deploy

Detailed Results

Status: Scored

Patient ID: AUTO_GEN_2CE6BFBC

Extracted Characters: 1199

Patient Inputs

Age: 25

BMI: 22.5

BMI Category: Normal

Smoker: No

Extracted Diabetes / Heart Indicators

	Indicator	Value	Unit	Reference	Flag
0	glucose	132	mg/dL	70–99	High
1	hba1c	6.7	%	< 5.7	High
2	cholesterol_total	235	mg/dL	< 200	High
3	ldl	160	mg/dL	< 100	High
4	hdl	38	mg/dL	> 40	Low
5	triglycerides	210	mg/dL	< 150	High
6	systolic_bp	145	mmHg	< 120	High
7	diastolic_bp	92	mmHg	< 80	High